

Aura AI – AI Unified Radiology Assistant

A Modular Flask-Based Web Application for AI-Powered Cancer Detection in Histopathology Images

Introduction

Aura AI – Unified Radiology Assistant is a next-generation, AI-powered diagnostic support system designed to assist pathologists, oncology researchers, and medical students in the rapid preliminary assessment of histopathology images. Leveraging deep learning (PyTorch + ResNet18) and explainable AI (Grad-CAM), Aura AI provides both accurate predictions and transparent visual reasoning—offering insights into why the model makes a particular decision.

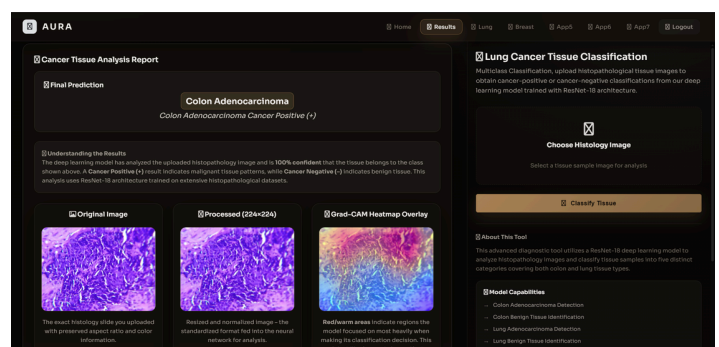
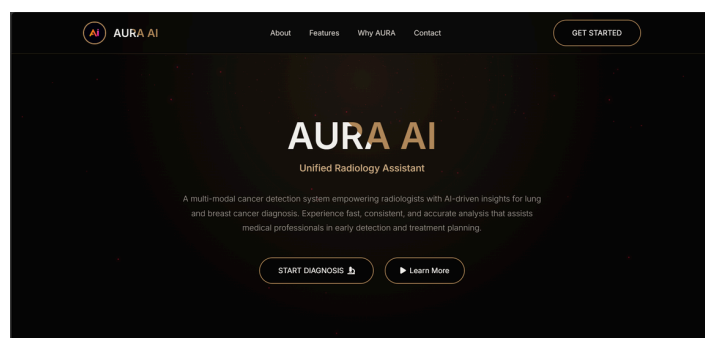
Purpose

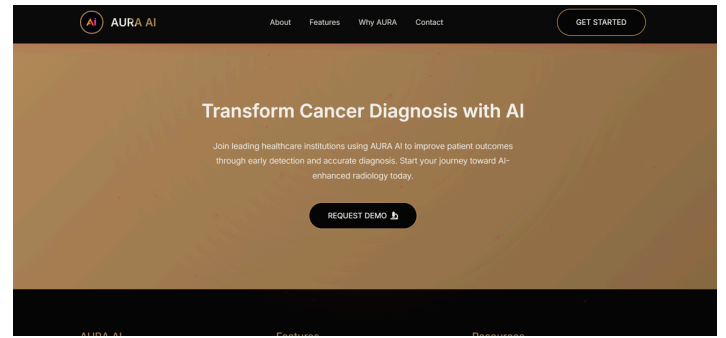
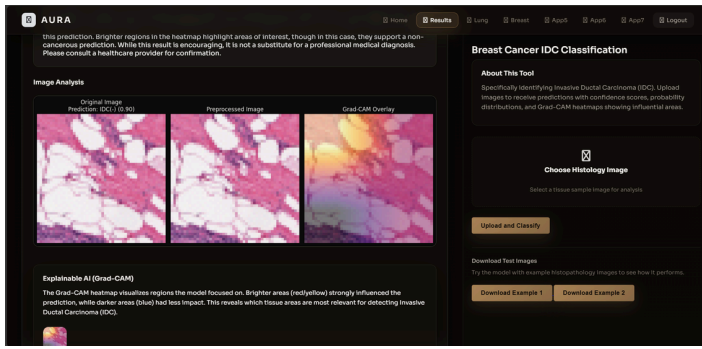
Aura AI is an advanced, user-centric histopathology analysis assistant built to support fast, reliable, and explainable assessment of cancerous tissues. The system streamlines preliminary diagnostic screening for two major cancer families—Breast Cancer and Lung & Colon Cancer—while combining deep learning, pathology-aligned workflows, and explainable AI to create a powerful tool for researchers, clinicians, and medical students.

Previous Works

Developed as a modular Flask application using the Application Factory + Blueprint pattern, the platform is scalable, maintainable, and easily extendable with new cancer models or diagnostic modules. Aura AI currently supports:

- Breast Cancer – IDC Detection (2-class)
- Lung & Colon Cancer – 5-class tissue classification





Breast Cancer (App4 — Primary Module)

The breast cancer module serves as one of the platform’s core diagnostic components. It performs binary classification to differentiate between:

Deep Learning Model we use Restnet 18

- IDC+ (Invasive Ductal Carcinoma Positive)
- IDC– (Invasive Ductal Carcinoma Negative)

To enhance user understanding and diagnostic confidence, each prediction is paired with:

- Grad-CAM heatmaps that highlight regions most influential to the model’s decision
- Clinical interpretation text that contextualizes the prediction
- Multi-image / batch processing, enabling users to screen multiple slides efficiently

This module is primarily designed for rapid IDC screening and educational visualization.

Lung & Colon Cancer (App3)

The Lung & Colon module provides a more diverse classification system, offering 5-class tissue categorization to help distinguish between malignant and benign cases across two organ systems:

Deep Learning Model we use Restnet 18

- Lung Adenocarcinoma
- Lung Squamous Cell Carcinoma
- Lung Benign
- Colon Adenocarcinoma
- Colon Benign

Each prediction is supported by visual analytics, including:

- Probability bar charts for intuitive comparison of model confidence across all classes

- Grad-CAM heatmaps that reveal high-impact tissue regions

This broader classification supports research use cases where tissue differentiation plays a critical role.

Explainable AI (XAI) Integration

Custom-built Grad-CAM implementation from scratch (no black-box models). Generates high-resolution, color-coded heatmaps superimposed on original microscopy images, enabling medical professionals to understand and trust every prediction.

Added Feature

Previously, we worked with lung and breast cancer histopathology images. We have now transitioned to MRI data and developed a brain tumor classification model using ResNet-101.

Model Training

This project presents a brain tumor classification system using [MRI images](#), built on a ResNet-101 deep learning architecture with transfer learning and explainable AI. The model classifies MRI scans into three categories: brain glioma, brain meningioma, and brain tumor, following a carefully designed end-to-end pipeline including preprocessing, augmentation, training, fine-tuning, and interpretation.

The backbone network is a pretrained ResNet-101 (ImageNet weights), chosen for its depth and strong feature-extraction capability. The complete model contains 45.8 million parameters (~174.8 MB). To balance performance and computational efficiency, only ~1.05 million parameters (4.01 MB) were made trainable during the initial training phase, while ~42.66 million parameters (162.7 MB) remained frozen to preserve learned visual representations. An additional ~2.10 million parameters (8.02 MB) were allocated to optimizer states during training.

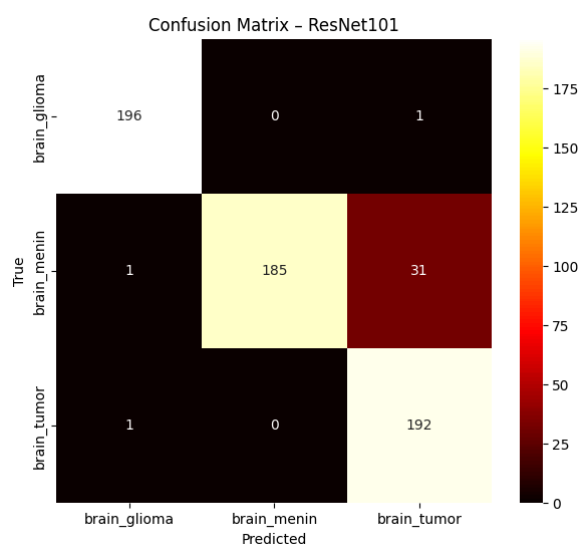
After convergence, selective fine-tuning of deeper convolutional layers further improved performance, resulting in ~96% validation accuracy and ~94% test accuracy, with strong class-wise precision and recall.

To ensure transparency and trustworthiness, the system integrates a custom Grad-CAM implementation, enabling visual explanation of model attention by highlighting MRI regions most influential to each prediction.

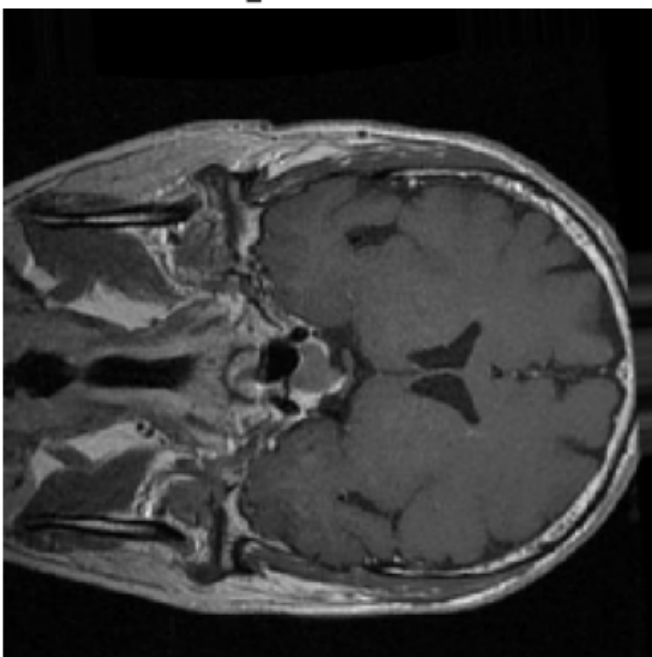
This project demonstrates effective use of large-scale pretrained models, parameter-efficient training, medical image classification, and explainable AI, marking a clear progression from earlier histopathology-based cancer analysis to MRI-based brain tumor detection.

For model training we use Kaggle T4 GPU

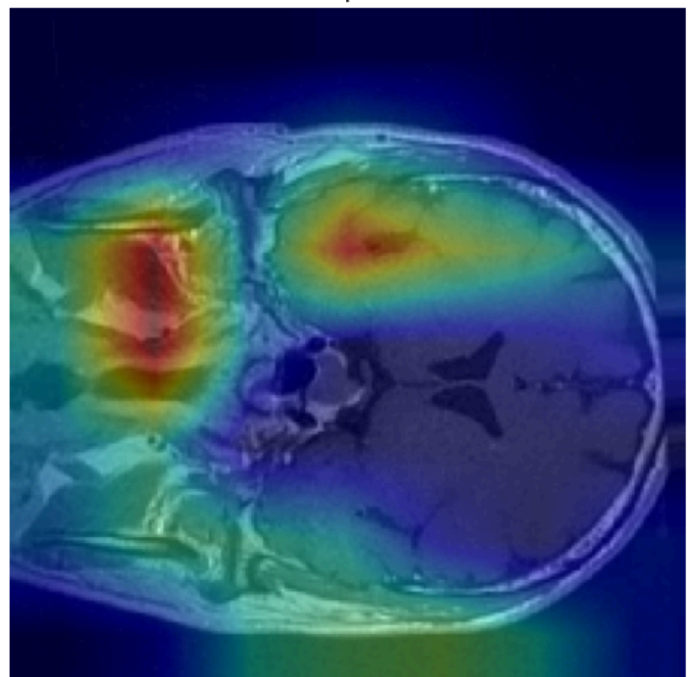
Note Book Link [Brain Cancer - MRI CLASSIFICATION](#)



brain_tumor (99.96%)



Grad-CAM Heatmap (Model Attention)



Back-end Implementation

The trained ResNet-101 brain tumor MRI model was integrated into the main Aura AI codebase as a new, independent Flask Blueprint (app5). This module follows the same modular architecture as existing cancer detection apps, with clearly separated routes, initialization, and templates, ensuring consistency, scalability, and maintainability across the platform.

The model is loaded once at application startup and exposed through dedicated inference routes, supporting MRI image upload, prediction, and explainable visualization. With this addition, Aura AI now supports histopathology-based cancer analysis (lung, colon, breast) as well as MRI-based brain tumor classification within a unified, production-ready system.

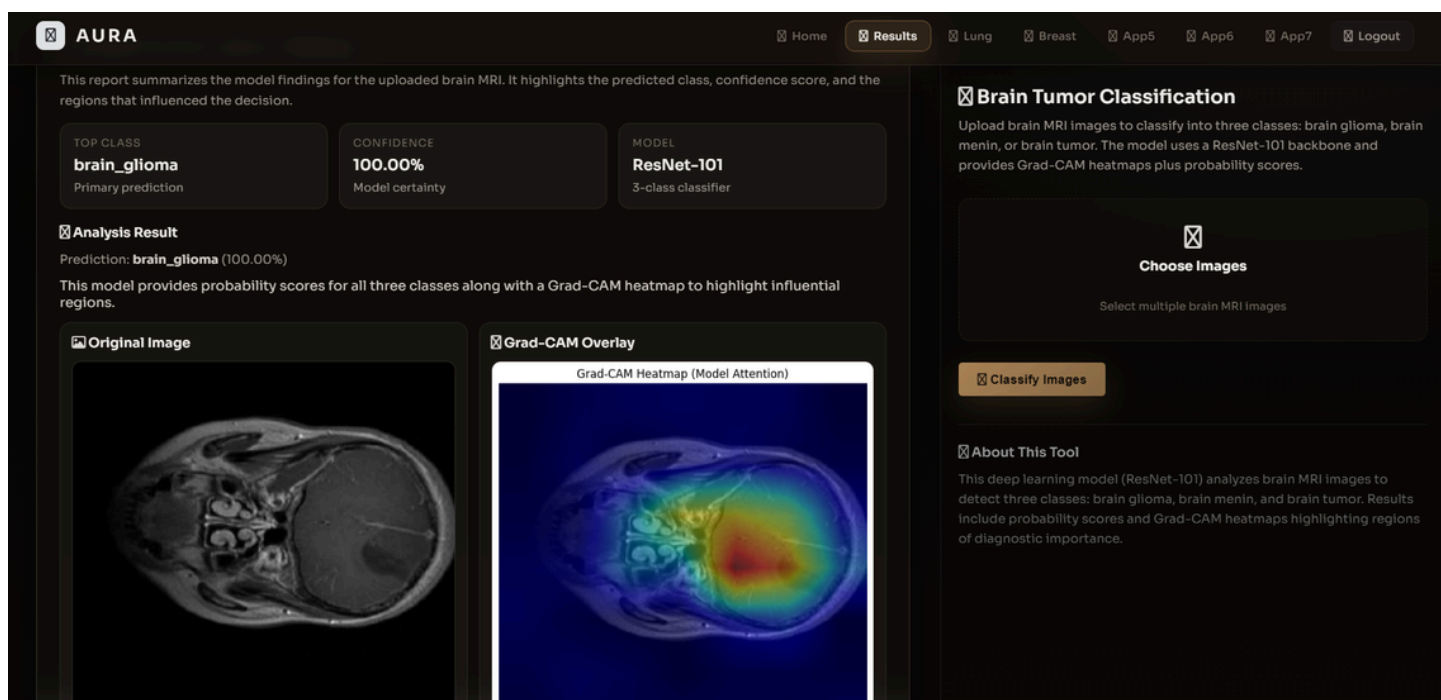
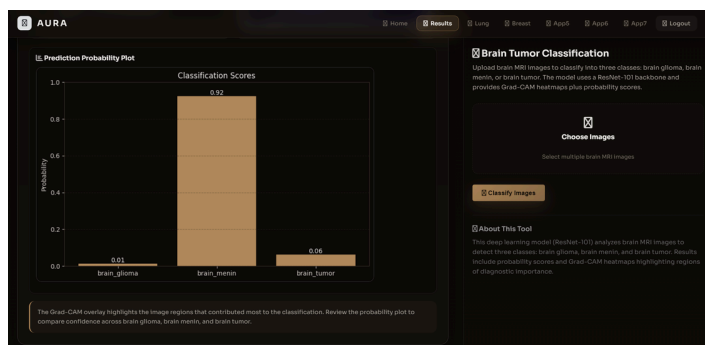
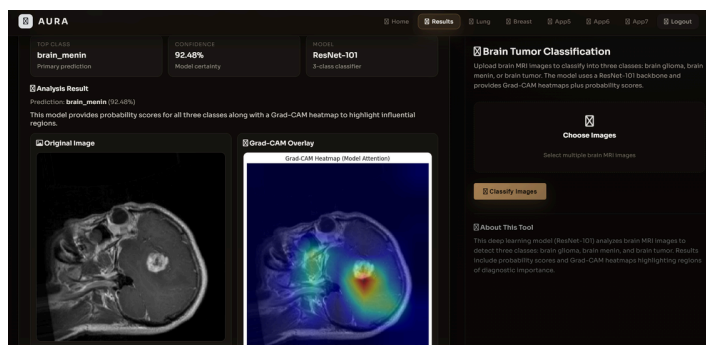
The entire application is Dockerized, enabling reproducible builds, environment isolation, and seamless deployment across development and production environment

Updated codebase

```
aura-ai/
├── app1/ # Home / Dashboard (post-login)
│   ├── templates/home.html
│   ├── routes.py
│   └── __init__.py
├── app2/ # Authentication Blueprint
│   ├── templates/
│   │   ├── login.html
│   │   ├── signup.html
│   │   └── verify.html
│   ├── routes.py
│   ├── models.py # User model + database schema
│   └── __init__.py
├── app3/ # Lung & Colon Cancer (5-class)
│   ├── templates/home2.html
│   ├── routes.py
│   ├── resnet18_model_001.pth
│   └── __init__.py
├── app4/ # Breast Cancer IDC Detection
│   ├── templates/home3.html
│   ├── routes.py
│   ├── static/uploads/ # Grad-CAM outputs
│   ├── breast_cancer_cnn_model_updated.pth
│   └── __init__.py
├── app5/ # Brain Tumor MRI Classification (NEW)
│   ├── templates/home4.html
│   ├── routes.py # MRI upload, prediction, Grad-CAM
│   ├── brain_tumor_resnet101_finetuned_v00.3.keras
│   └── __init__.py
├── instance/ # SQLite DB (gitignored)
├── images/ # README / UI screenshots
├── static/ # Global CSS / JS
├── templates/ # Global base.html
├── Dockerfile # Container definition
├── docker-compose.yml # Multi-service setup (optional)
├── .env # Environment variables
├── .gitignore
├── requirements.txt
├── run.py # Application entry point
└── README.md
```

Tech Stack	
Layer	Technology
Backend Framework	Flask 3.x, Flask-SQLAlchemy, Flask-Migrate, Flask-Mail
Frontend	HTML5, CSS3, Jinja2 templating, Vanilla JS
Deep Learning	PyTorch 2.x, torchvision, OpenCV, NumPy, Matplotlib/Seaborn
Database	SQLite (dev) → easy migration to PostgreSQL/MySQL
Authentication	Flask sessions + bcrypt hashing
Explainability	Custom Grad-CAM implementation
Deployment Ready	.env support, Gunicorn-compatible, requirements.txt

This is our updated project featuring a clean, dark-themed UI that classifies brain cancer from user-uploaded MRI images, while displaying performance metric graphs and Grad-CAM visualizations for model interpretability.



Conclusion

Aura AI has evolved into a modular, production-oriented medical AI platform that demonstrates how deep learning models can be responsibly designed, integrated, and explained within a real-world system. What began as histopathology-based cancer analysis for lung, colon, and breast tissues has now been successfully extended to MRI-based brain tumor classification using a ResNet-101 architecture, highlighting the platform's flexibility and scalability.

The project combines parameter-efficient transfer learning, selective fine-tuning, and custom explainable AI (Grad-CAM) to deliver high-performing models while maintaining transparency and trust. With clear separation of concerns through Flask Blueprints, consistent UI design, and Dockerized deployment, Aura AI moves beyond experimental notebooks into a deployable, maintainable application.

By integrating multiple cancer detection pipelines—each with its own data modality, model, and interpretability layer—Aura AI demonstrates a unified approach to medical image analysis that balances performance, interpretability, and system design. While intended strictly for research and

educational use, the project serves as a strong technical foundation for future expansion into additional imaging modalities, advanced reporting, and large-scale deployment.

Overall, Aura AI reflects a complete end-to-end workflow—from data and model development to UI, explainability, and deployment—showcasing not only machine learning capability, but also engineering discipline and product-level thinking.